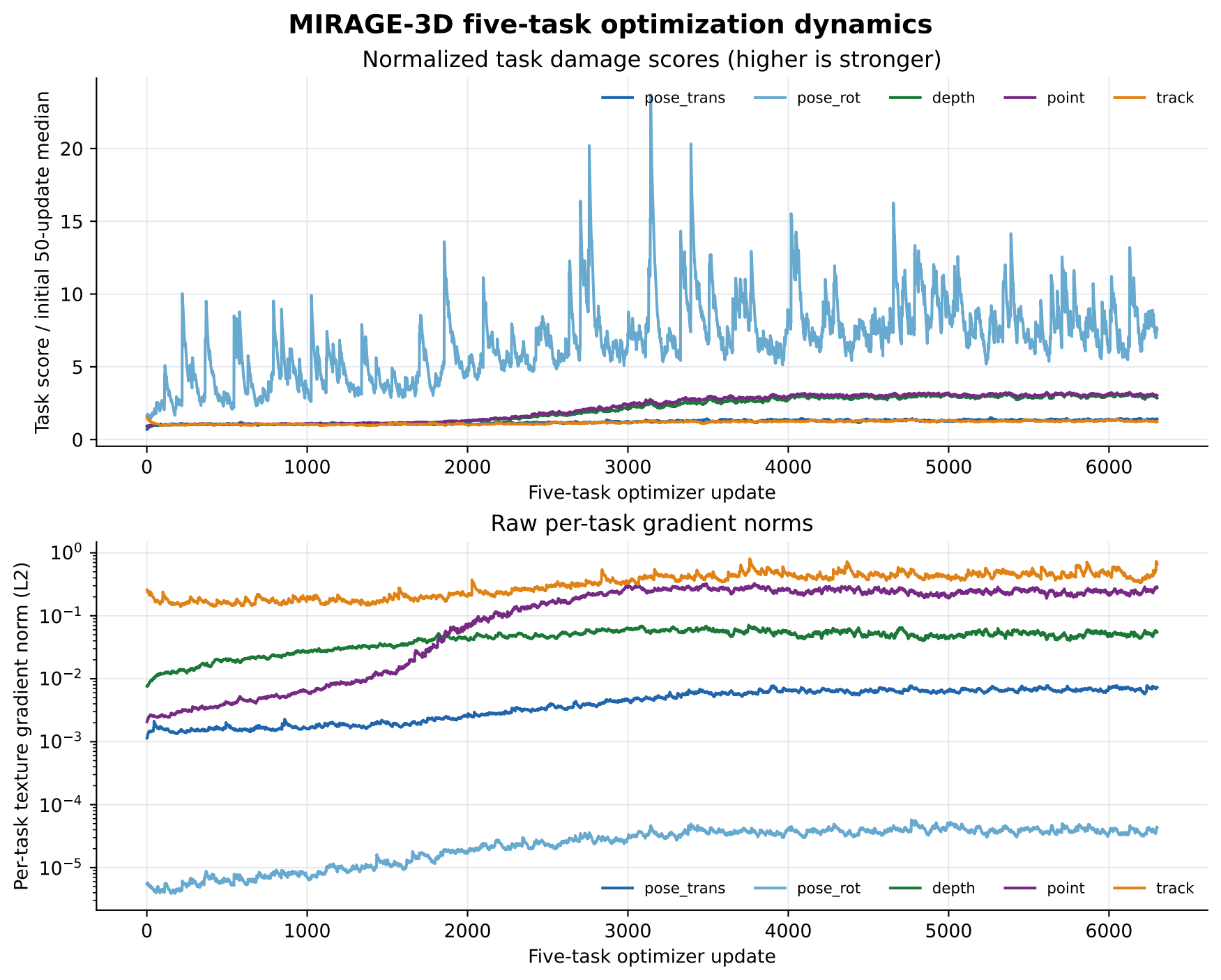

MIRAGE-3D five-task optimization dynamics
Normalized task damage scores (higher is stronger)
pose_trans
pose_rot
depth
point
track
Task score / initial 50-update median
0
5
10
15
20
0
1000
2000
3000
4000
5000
6000
Five-task optimizer update
Raw per-task gradient norms
Per-task texture gradient norm (L2)
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
pose_trans
pose_rot
depth
point
track
0
1000
2000
3000
4000
5000
6000
Five-task optimizer update